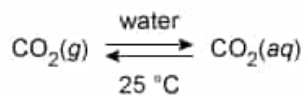
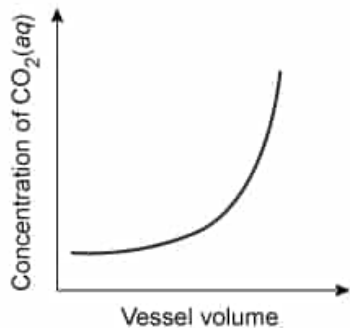


A sealed reaction vessel with a movable piston contains $\text{CO}_2(g)$ in equilibrium with $\text{H}_2\text{O}(l)$ at 25°C , as shown by the dissolution equation below.



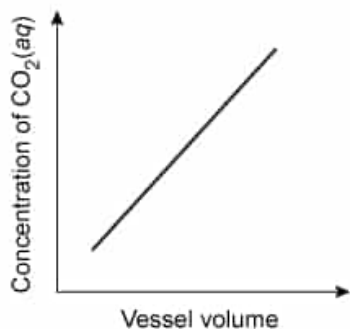
If the volume of the reaction vessel is increased, which graph best describes the effect of volume on the concentration of dissolved $\text{CO}_2(aq)$?

☐ A.



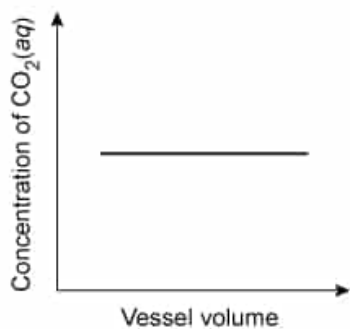
[14%]

☐ B.



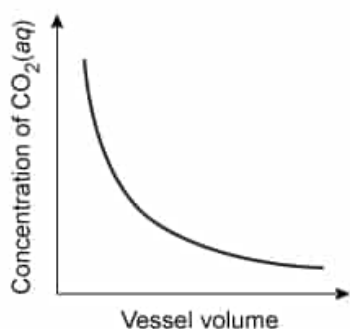
[16%]

☐ C.



[11%]

✓ ☒ D.

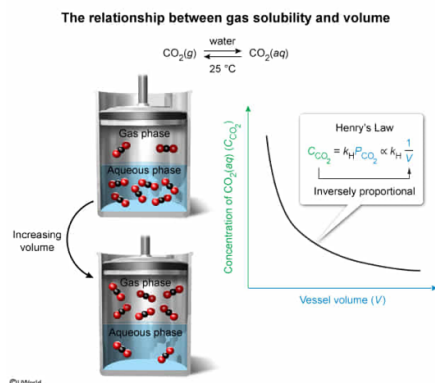


[57%]

Correct

57% answered correctly

Explanation:



At a fixed temperature, the amount of gas dissolved in a particular solvent is **directly proportional** to the equilibrium partial pressure of that gas present above the solvent, as described by **Henry's law of solubility**:

$$C = k_H P_{\text{gas}}$$

where C is the concentration (or solubility) of the dissolved gas, k_H is the solubility coefficient specific to the gas and solvent at a given temperature, and P_{gas} is the **partial pressure** of the gas at equilibrium. If the pressure exerted by the gas on the solvent **increases** (or decreases), the amount of gas dissolved in the solvent will also **increase** (or decrease) by the same factor.

The question states that the volume of the reaction vessel containing an equilibrium mixture of dissolved $\text{CO}_2(\text{aq})$ and $\text{CO}_2(\text{g})$ is increased. According to **Boyle's law**, volume and pressure are **inversely proportional**:

$$P \propto \frac{1}{V}$$

If volume **increases** (or decreases), pressure **decreases** (or increases) by the same factor. Substituting this relationship into Henry's law yields:

$$C \propto k_H \frac{1}{V}$$

Therefore, as the volume of the reaction vessel increases (ie, the partial pressure of $\text{CO}_2(\text{g})$ decreases), the concentration of dissolved $\text{CO}_2(\text{aq})$ decreases (ie, inversely proportional to volume), as depicted by the graph in Choice D.

(Choices A and B) These graphs incorrectly show that the concentration of $\text{CO}_2(\text{aq})$ increases with increasing volume.

(Choice B) This graph incorrectly shows the relationship between volume and $\text{CO}_2(\text{aq})$ concentration to be *directly* proportional. The *partial pressure* of $\text{CO}_2(\text{g})$ is directly proportional to the concentration of $\text{CO}_2(\text{aq})$, not the volume of the vessel.

(Choice C) This graph indicates that the concentration of $\text{CO}_2(\text{aq})$ is not affected by the volume of the reaction vessel (ie, the partial pressure of $\text{CO}_2(\text{g})$), which cannot be true according to Henry's law.

Educational objective:

According to Henry's law, the solubility of a gas in a liquid is directly proportional to the partial pressure of the gas over the liquid. Because pressure is inversely proportional to volume (Boyle's law), gas solubility is inversely proportional to the volume of the reaction vessel.

Time spent: 0 sec
Subject:
General Chemistry

QID: 403180
System:
Solutions and Electrochemistry